

Exercise 1.2: Box Detection using RANSAC and Image Processing Limitations and Potential Improvements

Rahul Sawhney

Chirag Sonani

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1 Introduction

In this document, we discuss the limitations of our box detection algorithm implemented for Exercise 1.1 and suggest potential improvements for more robust performance. Our method uses RANSAC to identify planes in point cloud data, applies morphological operations to enhance segmentation quality, and calculates box dimensions based on the detected floor and box top planes. While our implementation successfully measures box dimensions across the provided test cases, we identified several limitations that affect reliability and accuracy, particularly in more challenging scenarios.

2 Limitations of the Current Implementation

2.1 RANSAC Plane Detection Constraints

During our implementation and testing process, we identified these limitations in our RANSAC-based approach:

- **Largest plane assumption:** Our algorithm assumes the floor corresponds to the largest plane in the scene. This assumption would fail if another large planar surface (like a wall) dominates the scene or if the floor is partially occluded. We observed this could potentially be an issue when examining `example4kinect`, where background elements appear in the scene.
- **Fixed threshold parameters:** We used fixed inlier thresholds that worked well for the test cases but would likely fail with different sensor noise levels. When testing various thresholds during development, we found that small changes in these values significantly impacted segmentation quality.
- **Noise sensitivity:** This limitation became evident in `example3kinect`, where sensor noise produced irregular edges in the box mask. The rough top edge of the box in Figure 3 of `example3kinect` demonstrates how noise can affect dimension calculations.

2.2 Morphological Processing Limitations

Our implementation's morphological operations have the following constraints:

- **Non-adaptive kernel size:** After experimenting with different kernel sizes, we settled on a 3×3 kernel for both erosion and dilation operations. While this works for the test cases, it doesn't adapt to the scale of objects in the scene. We noticed that larger objects would benefit from larger kernels (5×5 or 7×7), while smaller details require smaller kernels.
- **Detail preservation:** We observed that some fine details along edges were lost during morphological operations. This is visible in the slightly rounded corners in `example1kinect`'s floor mask, potentially affecting the accuracy of box boundary detection.

2.3 Box Orientation and Dimension Estimation

When examining our results across different examples, we identified these challenges:

- **Orientation assumptions:** Our algorithm doesn't explicitly account for box orientation. Instead, we calculate dimensions based on min/max coordinates, which works reasonably well for boxes aligned roughly with image axes but leads to inaccuracies for rotated boxes. This is evident when comparing results between `example1kinect` and `example3kinect`, where the box is positioned differently.
- **Corner detection precision:** The visualizations in Figure 4 for each example show that our corner detection relies on simple min/max operations on the box mask. This approach is susceptible to noise and irregularities, as seen in the box corners of `example3kinect`.
- **Performance validation:** Without ground truth measurements for the boxes, we have no objective way to evaluate the accuracy of our dimension estimates. We attempted to visually verify the proportions but lacked quantitative validation.

3 Proposed Improvements

3.1 Enhanced RANSAC Implementation

Based on our observations, we propose these improvements to the RANSAC implementation:

- **Model confidence criteria:** We could implement early stopping based on the stability of detected plane parameters across iterations rather than relying solely on iteration count or inlier ratio. This would improve efficiency by terminating the search when a consistently stable model is found.
- **Adaptive thresholds:** A more robust approach would dynamically adjust inlier thresholds based on the estimated noise levels in the input data. We could implement this by analyzing the distance distribution of points to their nearest neighbors.
- **Multi-plane detection:** Rather than our sequential approach (detecting floor first, then box), a more sophisticated implementation could simultaneously detect multiple planes and classify them based on orientation and relative position.

3.2 Advanced Geometric Analysis

To address the orientation and precision issues we encountered:

- **Principal Component Analysis (PCA):** By applying PCA to the detected box points, we could determine the principal axes of the box regardless of its orientation. This would solve the rotation issue we observed in `example3kinect` and `example4kinect`, where the box is not aligned with the image axes.
- **Edge and corner refinement:** Sub-pixel edge and corner detection techniques would improve dimension measurement precision, particularly for boxes with irregular boundaries as seen in `example3kinect`.
- **3D model fitting:** Instead of treating the box as separate planes, we could implement a complete 3D box model fitting algorithm that incorporates all visible sides of the box simultaneously, providing more coherent dimension estimates.

3.3 Multi-view and Temporal Integration

For future applications with access to multiple viewpoints or temporal sequences:

- **Multi-view fusion:** Combining measurements from different camera angles would significantly improve accuracy. In our testing, we noticed that certain box features were more visible from particular angles, suggesting that integrating multiple views would provide more complete information.
- **Temporal filtering:** For video sequences, we could implement Kalman filtering to stabilize dimension estimates over time, reducing the impact of frame-to-frame noise that we observed in our static examples.
- **Confidence-weighted integration:** We could assign confidence scores to measurements based on viewing angle and distance, allowing for more intelligent integration of information from multiple sources.

4 Conclusion

Our box detection implementation successfully identifies and measures boxes in the provided test cases, demonstrating the effectiveness of our RANSAC-based approach. However, through our testing and analysis, we identified limitations related to plane detection assumptions, morphological processing, and box orientation handling.

The improvements we've proposed—enhanced RANSAC implementation with adaptive thresholds, geometric analysis using PCA, and multi-view integration—would address these limitations and provide a more robust solution. While implementing these improvements would require additional computational resources, they would significantly enhance reliability across a wider range of scenarios.

For future work, we would particularly focus on implementing PCA-based orientation detection and adaptive morphological operations, as these would address the most significant limitations we observed in our current implementation.